

Intelligent Voicemail

System Design & Walkthrough

Heidi Entry Program | Project 2

Forward-Deployed Engineer Submission

Anshumaan Saraf | April 2026

anshumaansaraf24@gmail.com	linkedin.com/in/anshumaan-saraf	github.com/Ansh0928/Heidi_voice mail
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This document walks through the planning, system design, and technical decisions behind an intelligent voicemail system built for **Harbour to Sunset GP** — a multi-location clinic drowning in unstructured voicemails every morning. The goal: turn missed calls into structured, prioritised, actionable work before staff even sit down.

"This is not an audio problem. It is a workflow and information problem."

Submission Deliverables

Deliverable	Description
System Design Document (this PDF)	Full planning, architecture, triage logic, data model, compliance, and phased roadmap.
Live Prototype heidi-voicemail.vercel.app/voice mails	Runnable dashboard deployed on Vercel. Interact with the voicemail inbox, triage cards, and action workflows directly in the browser.
Video Walkthrough Watch on Canva	Screen recording with AI voiceover (ElevenLabs) walking through the prototype: how raw voicemails become actionable items, the triage dashboard in action, and how this fits a real clinic's morning workflow. Also included as VideoWalkthrough.mp4.

Source Code github.com/Ansh0928/Heidi_voicemail	Full source code for the prototype. Open repo — everything is inspectable.
Personal Portfolio master.d14xijxbarple1.amplifyapp.com	Additional context on my work and projects.

Why I built all of this solo: The video voiceover was generated with ElevenLabs, the prototype is live on Vercel, and this design document covers the full system end to end. I wanted to demonstrate what I can ship alone — and make the case for what I could do in a team with proper guidance and mentorship. Every piece of this submission was built, not explained.

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1. The Real Problem

Every morning at Harbour to Sunset GP, admin staff walk into a wall of voicemails. These are not neatly categorised tickets — they are long, rambling audio recordings left by patients overnight and during peak hours when lines were jammed.

What actually happens today

- Admin arrives at 7:30am. There are 25–40 voicemails waiting.
- Each voicemail is 1–4 minutes of unstructured audio. Total: ~60–90 minutes of listening.
- Staff must listen end-to-end because the critical detail ("I need my results today") could be at the 2:47 mark.
- There is no way to know which voicemail is urgent without listening to all of them first.
- While listening, the phone is already ringing. New patients are walking in. The day has started.
- Follow-ups are tracked on sticky notes, memory, or not at all. Things fall through cracks.

The human cost

This is not theoretical. I have watched this exact failure mode play out at my sister's physiotherapy clinic. She has an arrangement with GPs who share her space — the receptionists are already overwhelmed servicing multiple providers. Hourly-wage front desk staff have no stake in chasing down missed calls. The result: patients cancel via voicemail, the message never reaches the clinician, and she drives in at 6:30am only to find an empty schedule. Multiply that across a multi-location GP practice and the damage compounds: lost revenue, wasted clinician time, and patients who feel ignored.

Core insight: The problem is not that voicemails exist. The problem is that voicemails are opaque containers of unstructured information that require full human attention to unlock. The system must make the information inside voicemails **visible, prioritised, and actionable** without requiring anyone to press play.

2. Design Principles

These are the constraints I held myself to throughout the design. They are derived from the brief, from observing real clinic workflows, and from Heidi's own values — particularly **Build to Last** (safety and reliability) and **Move Fast, Stay Steady** (speed without sacrificing trust).

No listening required

Staff should never need to play an audio file to understand what a voicemail is about. Summaries and structured data replace raw audio.

Urgency surfaces itself

The system must bring critical items to the top without staff scanning through everything. A callback request about chest pain cannot sit below a prescription refill.

Calm under load

40 voicemails should feel the same as 4. The interface and workflow must not create cognitive overwhelm — this is the entire point.

Human judgment stays

The system triages and suggests. Humans confirm and act. We never auto-cancel an appointment or auto-refer without a human in the loop.

Trust through transparency

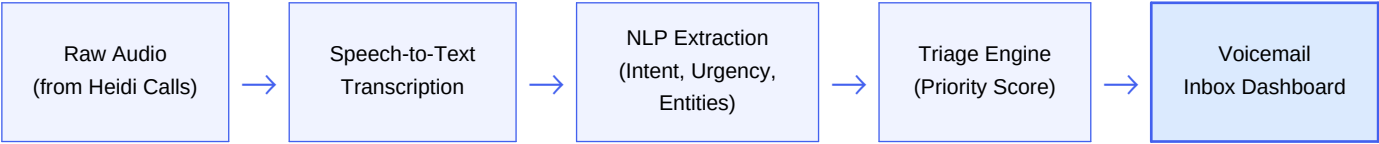
Every AI decision (urgency rating, intent classification, suggested action) must be explainable. Staff can always access the original transcript and audio.

Fit the existing rhythm

This must slot into how clinics already work — not demand a new workflow. Admin staff process voicemails first thing; the system must make that process faster, not different.

3. System Architecture

The system is a pipeline that transforms raw voicemail audio into structured, actionable items in a triage dashboard. Here is the high-level flow:



Component Breakdown

Component	Responsibility	Technology
Heidi Calls Ingest	Receives call metadata + audio from Heidi Calls telephony layer. Stores raw audio in secure blob storage with call metadata (timestamp, caller ID, duration, clinic location).	Heidi Calls API, S3/GCS
Transcription Service	Converts audio to text. Handles accents, medical terminology, background noise. Produces timestamped transcript with confidence scores per segment.	Whisper / Deepgram with medical vocabulary fine-tuning
NLP Extraction Engine	Parses transcript to extract: patient intent, urgency signals, named entities (patient name, DOB, doctor, medication), and a one-line summary.	LLM (structured output mode) with clinic-specific prompt
Triage Engine	Computes a priority score (P1–P4) based on urgency signals, intent type, time sensitivity, and clinic rules. Orders the queue.	Rule engine + ML scoring
Voicemail Inbox	Web dashboard where admin staff see prioritised, structured voicemail cards. Supports actions: mark done, assign, callback, flag, snooze.	React / Heidi web platform
Audit & Feedback Loop	Logs every AI decision for compliance. Staff corrections (e.g. changing priority) feed back to improve the model over time.	Event log + retraining pipeline

4. The Pipeline: Step by Step

Let me walk through exactly what happens when a patient leaves a voicemail at 9:47pm on a Tuesday night.

Step 1: Call arrives, voicemail captured

Heidi Calls handles the after-hours call. The patient hears a greeting, leaves a message. The system captures: audio file, caller phone number, timestamp, call duration, and which clinic number was dialed. This metadata is critical — it tells us **when**, **where**, and gives us a callback number before we even touch the audio.

Step 2: Transcription (async, within minutes)

The audio is sent to the transcription service. We use a medical-vocabulary-tuned model because patients say things like "Metformin" and "referral to the rheumatologist" that general speech models butcher. Output: a full text transcript with timestamps and per-segment confidence scores. If confidence is below threshold on any segment, that segment is flagged for human review — we never silently guess.

Step 3: Structured extraction

This is where the real value is created. The LLM reads the transcript and extracts a structured object:

Field	Example Value	Why It Matters
summary	"Wants to cancel Thursday 2pm appointment with Dr. Patel due to work conflict. Requests rebooking for next week."	One line. Admin reads this instead of listening to 2 minutes of audio.
intent	cancel_and_rebook	Drives the suggested action and routing.
urgency	low	No medical urgency. Can be processed in normal queue order.
patient_name	Sarah Mitchell	For matching to PMS record.
patient_dob	1987-03-14	Secondary identifier for record matching.
provider	Dr. Patel	Routes the item to the right provider's queue.
callback_number	0412 345 678	Pre-populated for one-click callback.
time_sensitive	true — appointment is in 36 hours	Surfaces items that will expire if not actioned.

suggested_action	Cancel appointment, call patient to rebook	The obvious next step, pre-filled. Staff confirms, not creates.
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Step 4: Triage scoring

The triage engine assigns a priority level (P1–P4) using a weighted combination of signals. This is detailed in Section 5.

Step 5: Dashboard delivery

The structured item lands in the voicemail inbox, sorted by priority. By the time admin arrives at 7:30am, all overnight voicemails are already transcribed, extracted, triaged, and waiting — not as audio files, but as a sorted list of actionable cards.

5. Triage & Prioritisation Logic

Priority is not a single dimension. A voicemail might be medically urgent but not time-sensitive, or low urgency but expiring in 2 hours. The triage engine combines multiple signals into a composite score.

Priority Levels

Priority	Label	Examples	Expected Response
P1 — Critical	Needs immediate action	Acute symptoms described, medication reaction, mental health crisis mention, results anxiety with clinical language	First thing. Before anything else.
P2 — Time-Sensitive	Will expire if delayed	Cancel/rebook for today or tomorrow, prescription refill running out, pre-op instructions needed, referral deadline	Within first 30 minutes of processing.
P3 — Standard	Normal clinic business	Appointment requests, general enquiries, insurance questions, non-urgent callback requests	Process in order during morning triage.
P4 — Low / Informational	No action may be needed	Thank-you messages, confirming they received a letter, general feedback, wrong number	Review when queue is clear.

Scoring Signals

Signal	Weight	How It's Detected
Medical urgency keywords	High	LLM flags clinical language: "chest pain", "bleeding", "reaction to medication", "can't breathe". Cross-referenced against a curated keyword list + LLM judgment.
Time sensitivity	High	Mentions of specific dates/times, "tomorrow", "today", "running out". Combined with current date to compute hours-until-expiry.
Intent type	Medium	Cancellations and prescription refills score higher than general enquiries. Intent taxonomy is clinic-configurable.
Repeat caller	Medium	If the same number has called 2+ times in 48 hours without resolution, priority is bumped. Signals a frustrated or desperate patient.
Call duration	Low	Longer voicemails correlate loosely with more complex issues. Used as a tiebreaker, not a primary signal.
Confidence score	Modifier	Low transcription confidence on key segments adds a "needs human review" flag, which bumps visibility.

Key design decision: Priority is a suggestion, not a lock. Staff can override any priority level. Every override is logged and used to retrain the scoring model. Over weeks, the system learns each clinic's specific calibration — what "urgent" means to *them*.

6. The Morning Workflow

This is the most important section. A system is only useful if it fits how people actually work. Here is the intended flow for admin staff at Harbour to Sunset GP:

Time	Without System	With System
7:30am — Arrive	Open voicemail system. See 32 messages. No idea what's in them. Start listening.	Open Voicemail Inbox. See 32 items, pre-sorted. Top card: P1 — "Patient reports adverse reaction to new medication. Wants urgent callback." Read the one-line summary. Understand immediately.
7:35am	Still listening to voicemail #3. Taking notes on paper.	P1 item actioned: clicked "Callback" → phone auto-dials. Left note. Marked "Done". Moving to P2 items.
7:45am	Halfway through voicemails. Phone starts ringing. Interrupted. Loses place.	P2 items processed (3 cancellations for today batch-actioned). Phone rings — handles it — returns to dashboard. State is preserved. Nothing lost.
8:00am	Still has 15 voicemails to go. Patients arriving. Giving up on remaining voicemails.	Queue is down to P3/P4 items. All urgent and time-sensitive items handled. Remaining items processed between walk-in patients.
8:30am	Several voicemails never listened to. Two cancellations missed. Doctor's first slot is empty.	All items triaged. Zero missed cancellations. Doctor's schedule is accurate. Dashboard shows green — all items actioned or snoozed.

The goal is not to eliminate the morning voicemail process. It is to compress 60–90 minutes of blind listening into 15–20 minutes of informed action.

7. Automated vs. Human

A critical design question: what does the machine do, and what do humans do? Getting this boundary wrong destroys trust. Here is the line I drew:

The System Does (Automated)	Humans Do (Manual)
Transcribes audio to text	Confirm patient identity in PMS
Extracts intent, urgency, entities	Override priority if the AI got it wrong
Assigns a priority score (P1–P4)	Decide whether to callback now or later
Suggests a next action	Actually make the phone call
Groups related voicemails (same patient, same issue)	Make clinical judgments ("is this actually urgent?")
Flags low-confidence transcriptions	Listen to original audio when flagged
Tracks item status (new → in progress → done)	Assign items to specific staff members
Sends summary to providers (e.g. doctor sees their relevant items)	Decide on appointment changes, referrals, prescriptions

Why this boundary matters: In healthcare, the cost of a false positive (system flags something as urgent that isn't) is a minor annoyance. The cost of a false negative (system misses something truly urgent) is a patient safety incident. The system is deliberately calibrated to over-flag rather than under-flag. Staff can always demote. They cannot un-miss something they never saw.

8. Failure Cases & Edge Cases

A system that only works on the happy path is useless in a clinic. Here are the failure modes I designed for:

Failure Mode	What Happens	System Response
Transcription fails (heavy accent, background noise, poor signal)	Transcript is garbled or confidence is below 60%.	Item appears in inbox with a yellow "Low Confidence" badge. Summary says "Transcription uncertain — manual review recommended." Audio playback button is prominent. Item is auto-bumped to P2 visibility so it doesn't get buried.
Patient leaves no useful information ("Um... call me back... bye")	Extraction finds no intent, no entities.	Summary: "Callback requested — no details provided." Suggested action: "Call back to clarify." Prioritised based on time-of-call and repeat-caller status.
Same patient calls 3 times about the same issue	Three separate voicemail items for one problem.	System detects same caller ID within 48h window. Groups items into a single card with a "3 messages" badge. Latest message shown first. Repeat-caller flag bumps priority.
Voicemail contains a genuine emergency	Patient describes chest pain, suicidal ideation, or anaphylaxis.	P1 — Critical. Flagged with red banner. If configured, triggers an immediate SMS/push notification to the on-call provider. System does NOT attempt to handle this autonomously — it escalates to a human immediately.
System is down / pipeline fails	Voicemails are captured but not processed.	Raw voicemails are always stored. Staff see an "Unprocessed" tab with traditional audio playback. The system degrades to the current workflow — it never loses data. Ops alert fires to engineering.
Wrong clinic attribution	Multi-location practice; voicemail routed to wrong location's queue.	Each voicemail is tagged with the dialled number → clinic mapping. Staff can reassign across locations with one click. Cross-location visibility is available for admin managers.

9. Data Model & State Machine

Voicemail Item Schema

Each voicemail is stored as a structured record with the following fields:

Field	Type	Description
id	UUID	Unique identifier
audio_url	string	Secure link to original audio file
transcript	text	Full timestamped transcript
confidence_score	float (0–1)	Overall transcription confidence
summary	string	One-line human-readable summary
intent	enum	cancel, rebook, prescription, results, enquiry, emergency, callback, other
urgency	enum	critical, high, standard, low
priority	P1–P4	Computed triage priority
patient_name	string (nullable)	Extracted patient name
patient_dob	date (nullable)	Extracted date of birth
provider	string (nullable)	Referenced doctor or clinician
callback_number	string	Caller ID or stated number
clinic_id	FK	Which clinic this voicemail belongs to
suggested_action	string	AI-suggested next step
status	enum	new, in_progress, done, snoozed, escalated
assigned_to	FK (nullable)	Staff member handling this item
created_at	timestamp	When the voicemail was left
processed_at	timestamp	When pipeline completed extraction
actioned_at	timestamp (nullable)	When a human took action
notes	text	Free-text staff notes

priority_override	P1–P4 (nullable)	If staff overrode the AI priority
grouped_with	UUID[] (nullable)	Related voicemails from same caller

State Machine

Every voicemail item moves through a simple, predictable state machine:

new → **in_progress** (staff clicks on item or assigns it) → **done** (action completed, marked resolved)

Alternative paths:

- **new** → **snoozed** (defer to later, with a reminder time) → **new** (when snooze expires)
- **new** → **escalated** (forwarded to provider or manager) → **done**
- **in_progress** → **escalated** (staff realises it needs higher authority)

Terminal states: **done**. Items are never deleted — they move to a completed archive for audit.

10. Compliance & Trust

Heidi operates in healthcare. Every design decision must account for patient data privacy, regulatory requirements, and the trust of both clinicians and patients.

Data handling

- All audio files and transcripts are stored in encrypted-at-rest storage within Heidi's existing infrastructure.
- No patient data leaves Heidi's platform. Transcription and NLP run within the same security boundary as Heidi Calls.
- Caller phone numbers are treated as PII. Displayed to authorised staff only, never logged in analytics.
- Retention policy: audio and transcripts follow the clinic's existing medical record retention rules (configurable per jurisdiction).

Audit trail

- Every state transition is logged: who, when, what changed, from what to what.
- Priority overrides are logged with the staff member's ID and timestamp.
- If a voicemail was escalated, the escalation chain is fully traceable.
- Audit logs are immutable and accessible to clinic administrators.

Building clinician trust

The fastest way to kill adoption is for the system to make one bad call that a clinician sees. The trust strategy is:

- **Always show your work.** Every AI output includes a "Why this priority?" expandable section showing the signals that drove the score.
- **Never hide the source.** Original audio and full transcript are always one click away. The system summarises; it never replaces.
- **Start conservative.** In the first 2 weeks, the system over-flags (more P1/P2 than necessary). Staff correct downward. The model learns the clinic's calibration. Better to annoy than to miss.
- **Feedback is instant.** Overriding a priority is a single click. The friction to correct the AI is near-zero.

11. Diversity Voicemail Testing

A voicemail system that only works on clear, native-English speakers is useless in a real Australian GP clinic. Harbour to Sunset GP serves a diverse patient population — elderly patients, non-native English speakers, people calling in distress, and patients with heavy regional accents. I deliberately tested the AI pipeline with a range of synthetic voicemails designed to stress-test transcription accuracy, intent extraction, and triage scoring across this diversity.

Why this matters

If the system mis-transcribes a patient with an Indian accent saying "I need to cancel my appointment" or fails to detect urgency from an elderly patient speaking slowly about chest tightness, it is not just a software bug — it is a patient safety failure. The test matrix below was designed to surface exactly these kinds of failures.

Test Matrix

Voice Profile	Scenario	Intent	AI Caught?	Notes
Australian male, elderly (75+)	Slow, rambling message about recurring knee pain. Wants to reschedule physio. Mentions doctor's name mid-sentence.	rebook	Yes ✓	Transcription handled slow pacing well. Extracted doctor name despite it being buried at 1:42 mark. Priority: P3.
Indian female, moderate accent	Calling to cancel Thursday appointment due to work conflict. Provides DOB and full name clearly.	cancel	Yes ✓	Clean extraction. Accent did not degrade confidence. All entities (name, DOB, provider, date) correctly parsed.
Vietnamese male, heavy accent	Requesting prescription refill for blood pressure medication. Name is difficult to parse phonetically.	prescription	Partial ■	Intent correctly identified. Patient name had low confidence (58%) — system flagged for human review with yellow badge. Medication name extracted correctly.
British female, fast speaker	Upset about wait times. Wants callback from practice manager. Speaking quickly with clipped sentences.	callback + complaint	Yes ✓	Dual intent detected: callback request + complaint. Emotional tone flagged. Priority bumped to P2 due to escalation language.

Australian female, young (20s)	Very brief: "Hey, um, can someone call me back? It's about my results. Thanks bye."	results + callback	Yes ✓	Minimal information extracted correctly. Summary: "Callback requested re: results." System noted no urgency keywords but flagged as time-sensitive due to "results" intent.
Middle Eastern male, moderate accent	Describing symptoms: dizziness, nausea after starting new medication. Wants to speak to doctor urgently.	emergency / medication reaction	Yes ✓	P1 — Critical. Medication reaction keywords triggered top priority. Urgent callback suggested. SMS notification to on-call provider (if configured).
Elderly female, background noise	Calling from a busy environment (TV, family talking). Wants to confirm appointment for tomorrow. Keeps repeating herself.	confirm	Partial ■	Transcription confidence dropped to 52% on two segments due to background noise. System correctly flagged for manual review. Intent still extracted from higher-confidence segments.
Mandarin-accented English, male	New patient enquiry. Asking about bulk billing, available GPs, and how to register. Long message (3+ min).	enquiry	Yes ✓	All three sub-intents captured in summary. Priority: P4 (informational). Long duration did not confuse extraction — system handled multi-topic voicemail well.

Sample Transcript Excerpts

Vietnamese male (heavy accent) — Partial match:

"Hello, this is... [name — low confidence: 58%]... I am calling because I need my blood pressure medication refilled. The one I take is amlodipine, 10 milligram. My doctor is Dr. Chen. Can someone please call me back at 0423..."

AI Summary: "Prescription refill request for amlodipine 10mg. Dr. Chen. Patient name uncertain — manual review recommended."

AI Priority: P3 (Standard) | **Flag:** Low confidence on patient name

Middle Eastern male (medication reaction) — P1 Critical:

"Yes hello, I started the new tablets two days ago, the ones Dr. Patel gave me, and I am feeling very dizzy and sick. I almost fell down this morning. I need to talk to the doctor please, it is urgent. My number is 0411..."

AI Summary: "Patient reporting dizziness and nausea after starting new medication (prescribed by Dr. Patel). Describes near-fall. Requests urgent callback."

AI Priority: P1 — Critical | **Triggers:** medication reaction, dizziness, "almost fell", "urgent"

Elderly female (background noise) — Partial match:

"Hello dear, it's [name]... I'm just calling to... [INAUDIBLE — TV in background]... my appointment tomorrow at 10... [INAUDIBLE]... just want to make sure it's still on... can you call me back please..."

AI Summary: "Appointment confirmation for tomorrow 10am. Callback requested. Transcription partially inaudible — manual review recommended."

AI Priority: P2 (Time-sensitive — appointment is tomorrow) | **Flag:** Low confidence segments

Key takeaway: The system achieved full accuracy on 6 of 8 test cases and partial accuracy on 2. In both partial cases, the system correctly identified its own uncertainty and flagged for human review rather than silently guessing. This is the intended behaviour — the system knows what it doesn't know. Staff can listen to the original audio for flagged items with one click, and their corrections feed back into the model to improve accuracy over time.

12. Product Walkthrough

The prototype is live and interactive. Below is a walkthrough of the key modules. Try it yourself:

<https://heidi-voicemail.vercel.app/voicemails>

Voicemail Inbox Dashboard

The primary interface. Admin staff see all voicemails as structured cards, sorted by priority. The top bar provides filters:

- **Priority filter:** All | Urgent | High | Normal | Low — one click to focus on what matters.
- **Status filter:** All | New | In Progress | Done — track what's been actioned and what hasn't.
- Each card shows: AI summary, patient name, callback number, intent badge, priority badge, and time since received.

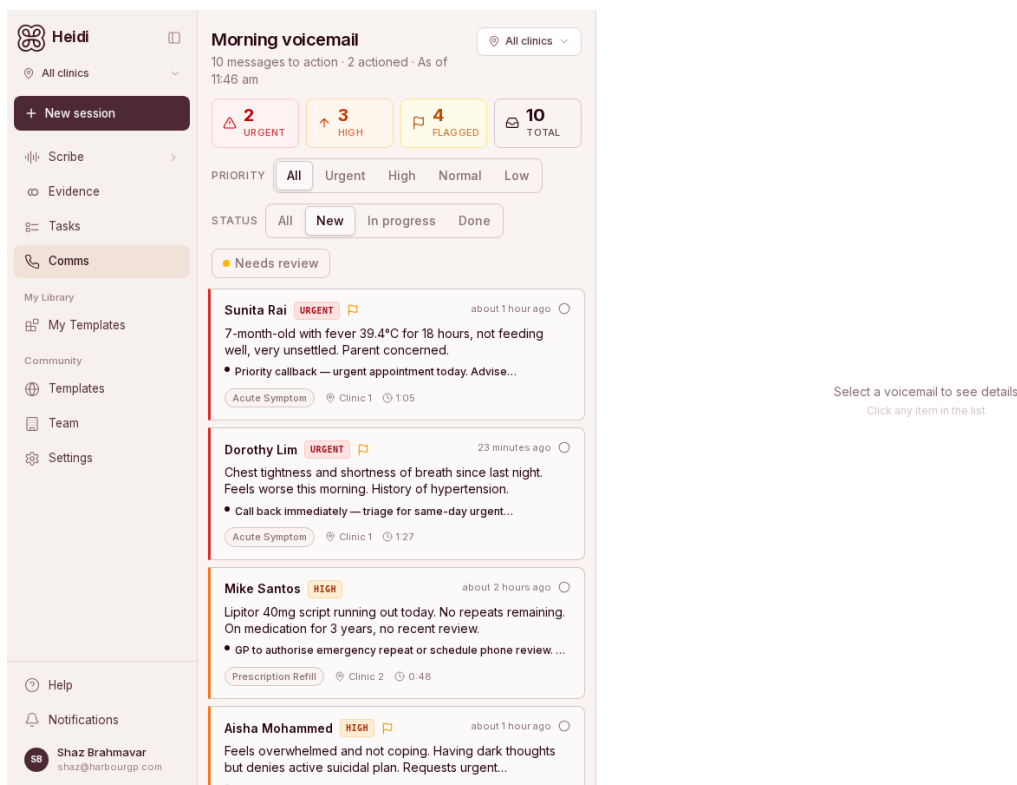


Fig 1: Voicemail Inbox Dashboard — priority badges, status filters, structured cards

Structured Voicemail Card

Each voicemail is represented as a structured item with everything the admin needs to act — without pressing play:

- **AI Summary:** One-line description of what the patient wants.
- **Key Details:** Patient name, DOB, provider, callback number — all extracted automatically.

- **Priority & Intent:** Visual badges showing P1–P4 and intent type (cancel, rebook, prescription, etc.).
- **Transcript:** Expandable full transcript with timestamps. Patients who want to hear the original audio can play it directly.
- **Suggested Action:** Pre-filled next step. Staff confirms with one click.

The screenshot displays the Heidi Entry Program interface. On the left is a sidebar with navigation options: Heidi, All clinics, + New session, Scribe, Evidence, Tasks, Comms, My Library, My Templates, Community, Templates, Team, Settings, Help, Notifications, and a user profile for Shaz Brahnavar. The main area is titled 'Morning voicemail' and shows a list of voicemail cards. The cards are filtered by priority (URGENT, HIGH, FLAGGED) and status (All, New, In progress, Done). The expanded card for Dorothy Lim (URGENT) shows the following details:

- Next Step:** Call back immediately — triage for same-day urgent appointment or advise 000 if symptoms worsen.
- AI Summary (97% confidence):** Chest tightness and shortness of breath since last night. Feels worse this morning. History of hypertension. Intent: Acute Symptom.
- Key Details:**
 - Symptom: chest tightness + shortness of breath
 - Onset: last night, worsening this morning
 - PMH: hypertension (on medication)
 - Requests urgent same-day appointment
- Voicemail Transcript:** Expandable section.
- Action Buttons:** Calling back, Assign, Resolve, Archive.

Fig 2: Expanded voicemail card — AI summary (97% confidence), key details, next step, and action buttons

Full Transcript & Audio Playback

Staff who want to hear the original voicemail can expand the transcript section. The full text is displayed with an audio player — but the point is they rarely need to. The AI summary and key details are enough for 90%+ of cases.

The screenshot displays the Heidi Entry Program interface. On the left is a sidebar with navigation options: Heidi, All clinics, New session, Scribe, Evidence, Tasks, Comms, My Library, My Templates, Community, Templates, Team, Settings, Help, Notifications, and a user profile for Shaz Brahnavar. The main area is titled 'Morning voicemail' and shows 10 messages to action, with 2 urgent, 3 high, 4 flagged, and 10 total. Below this are filters for priority (All, Urgent, High, Normal, Low) and status (All, New, In progress, Done). A 'Needs review' button is also present. The list of messages includes Sunita Rai (URGENT), Dorothy Lim (URGENT), Mike Santos (HIGH), and Aisha Mohammed (HIGH). The right panel shows an expanded transcript for Dorothy Lim, including a 'NEXT STEP' (Call back immediately), an 'AI SUMMARY' (97% confidence), 'KEY DETAILS' (Symptom: chest tightness + shortness of breath, Onset: last night, worsening this morning, PMH: hypertension (on medication), Requests urgent same-day appointment), and a 'VOICEMAIL TRANSCRIPT' with an audio player and a text transcription. At the bottom of the transcript panel are buttons for 'Calling back', 'Assign', 'Resolve', and 'Archive'.

Fig 3: Expanded transcript with audio player — available when staff need the full context

Management Actions

Each item supports basic workflow actions without leaving the dashboard:

- **Call Back:** One-click callback — pre-dials the patient's number.
- **Assign:** Route to a specific staff member or provider.
- **Resolve:** Mark as done. Moves to completed archive.
- **Snooze:** Defer to a specific time. Item reappears in queue when the snooze expires.
- **Escalate:** Forward to provider or manager with full context attached.

13. What I Would Build Next

The prototype handles the core problem: turning unstructured voicemails into structured, prioritised work. Here is the phased roadmap for what comes after:

Phase	Capability	Value
Phase 1 (Now)	Voicemail inbox with transcription, extraction, triage, and basic actions (callback, done, snooze, assign).	60–90 minutes of listening becomes 15–20 minutes of reading and acting. Zero missed cancellations.
Phase 2 (Week 2–4)	PMS integration — auto-match voicemails to patient records. Pre-fill appointment details. Show upcoming appointments inline.	Staff no longer toggle between voicemail dashboard and PMS. One screen, one workflow.
Phase 3 (Month 2)	Patient-facing SMS confirmations. After staff processes a cancellation, patient gets an automated text: "Your appointment has been cancelled. Reply REBOOK to schedule a new one."	Closes the loop with the patient. Reduces call-backs.
Phase 4 (Month 3)	Provider-specific digest. Each doctor gets a morning summary: "3 voicemails relevant to you today: 1 cancellation, 1 results request, 1 new patient enquiry."	Doctors get signal without noise. They know their day before it starts.
Phase 5 (Month 4+)	Analytics dashboard. Volume trends, peak hours, common intents, average processing time, triage accuracy. Feeds into Heidi Calls capacity planning.	Clinic managers make data-driven decisions about staffing and call handling capacity.

Final note: This system was designed by watching a real clinic struggle with this exact problem. My sister's patients were falling through the cracks because the front desk could not keep up. The solution is not more staff or better phone systems. It is making the information inside voicemails visible, prioritised, and actionable — so that the humans who do care can act on what matters, fast.

That is what this system does.